

IU International University of Applied Sciences

Cyber Security

Assignment: Task 1 – Use NLP techniques to analyze a collection of texts

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Table of Contents

1. Conception Phase	1
1.1 Data source	1
1.2 Data preprocessing.....	1
1.3 Text vectorization	1
1.4 Topic extraction	1
1.5 Python libraries	1
2. Development Phase/Reflection Phase	2
3. Finalization Phase	3
3.1 Abstract	3
3.2 Data Collection and Preprocessing.....	3
3.3 Vectorization Methods	3
3.4 Topic Modeling Techniques	4
3.5 Results and Evaluation	4
3.6 Reflection and Learning Outcomes.....	5
3.7 Limitations and Future Improvements.....	5
3.8 Conclusion	5

1. Conception Phase

1.1 Data source

For this project, an open dataset containing written user reviews is required to simulate large-scale written feedback similar to that received by public institutions. As a suitable proxy, a publicly available dataset of hotel reviews from TripAdvisor is selected. The dataset consists of unstructured textual reviews written by users, expressing evaluations, criticism, praise, and recurring concerns. The structure of the reviews is comparable to citizen feedback, as both consist of free-form text describing perceived issues, experiences, and expectations. The dataset includes 20 thousand entries, making manual analysis impractical and justifying the application of Natural Language Processing (NLP) techniques.

1.2 Data preprocessing

To obtain clean text, the data will undergo several preprocessing steps: conversion to lowercase, removal of punctuations, special characters, numbers, URLs, and stop words, as well as tokenization. Lemmatization will also be applied to reduce words to their base forms, improving semantic consistency.

1.3 Text vectorization

Two vectorization approaches will be used:

- Bag-of-Words / TF-IDF, which captures word frequency and importance.
- Word embeddings (e.g., Word2Vec or spaCy embeddings), which preserve semantic relationships between words.

1.4 Topic extraction

To identify prevalent topics, two methods will be applied:

- Latent Dirichlet Allocation (LDA) for probabilistic topic modeling.
- Non-negative Matrix Factorization (NMF) for matrix-based topic extraction.

1.5 Python libraries

The analysis will use Python libraries including pandas, NumPy, NLTK, spaCy, scikit-learn, and gensim.

2. Development Phase/Reflection Phase

In the development phase, the data analysis workflow defined in the conception phase was implemented using Python. The complete source code and documentation are hosted in a public GitHub repository: <https://github.com/EndritRamizi/Endrit>. A dedicated virtual environment was created using venv, and all required Python libraries were installed and exported to a requirements.txt file to ensure reproducibility.

The TripAdvisor hotel review dataset comprising over 20,000 entries was loaded with pandas and inspected for missing values and text consistency. To obtain clean input data, several preprocessing steps were applied, including lowercasing, removal of punctuation, URLs and numbers, stopword filtering, tokenization and normalization of tokens.

Two vectorization techniques were implemented and compared. TF-IDF representations were generated using scikit-learn to capture important words and phrases and provided highly interpretable features for topic modeling. In addition, semantic word embeddings were created by training a Word2Vec model on the corpus and averaging word vectors to produce document-level representations. While TF-IDF enabled transparent interpretation of topics, embeddings captured semantic similarity but were less suitable for explaining topic keywords.

To extract prevalent themes, two topic modeling techniques were applied. Latent Dirichlet Allocation (LDA) was trained on count vectors, and Non-Negative Matrix Factorization (NMF) was applied to TF-IDF matrices. Both approaches revealed recurring topics related to staff service, room quality, location and accessibility, general satisfaction, and resort facilities such as beaches, pools and food services. During implementation, challenges arose with library configuration and parameter tuning; these issues were resolved by consulting documentation and iteratively adjusting vectorizer thresholds and the number of topics.

Topic prevalence was quantified by computing the dominant topic for each review and exporting the resulting statistics to CSV files for further analysis. In the NMF model, room quality and related issues emerged as the most prevalent theme, dominating 25.5% of all reviews, followed by location and accessibility (22.6%) and resort amenities including food and pool areas (20.1%). In contrast, LDA ranked staff service and overall experience highest, accounting for 27.9% of reviews, while location remained consistently prominent (23.5%). The strong overlap between the two approaches indicates that accommodation conditions, staff service, location factors and resort facilities constitute the core themes within the corpus, increasing confidence in the robustness of the extracted topics.

3. Finalization Phase

3.1 Abstract

This project applied Natural Language Processing (NLP) techniques to extract prevalent themes from large collections of unstructured textual feedback in order to simulate how municipalities or public institutions could systematically analyze citizen complaints. Due to the lack of publicly available municipal complaint corpora, a large-scale dataset of TripAdvisor hotel reviews comprising over 20,000 entries was selected as a proxy. The dataset resembles citizen feedback in structure and content, as it contains free-form evaluations, criticism and recurring concerns.

The project followed a three-phase workflow consisting of conception, development and finalization. In the conception phase, preprocessing strategies, vectorization approaches and topic modeling techniques were defined in advance to ensure a transparent and reproducible analysis pipeline. The development phase implemented these concepts in Python, while the finalization phase evaluates the results, reflects on challenges encountered, and outlines potential improvements.

3.2 Data Collection and Preprocessing

The dataset was loaded using pandas and inspected for missing values and inconsistencies. To obtain clean textual input suitable for NLP analysis, several preprocessing steps were applied: conversion to lowercase, removal of punctuation, URLs and numbers, stopwords filtering, tokenization and normalization of tokens. These steps reduced noise and improved semantic consistency across the corpus.

All experiments were conducted in a dedicated Python virtual environment, and dependencies were exported in a requirements.txt file to ensure reproducibility. The full implementation and documentation were published in a public GitHub repository.

3.3 Vectorization Methods

Two different text representation techniques were implemented to compare their suitability for topic extraction. First, TF-IDF vectorization using scikit-learn transformed documents into sparse word-frequency matrices that emphasize informative terms and enable transparent interpretation of topics. Second, Word2Vec embeddings were trained directly on the corpus,

and document-level vectors were constructed by averaging individual word embeddings. While Word2Vec captured semantic similarity between documents, TF-IDF proved more interpretable when labeling and explaining topics to decision-makers.

3.4 Topic Modeling Techniques

To extract prevalent themes, two unsupervised topic modeling approaches were applied. Latent Dirichlet Allocation (LDA) was trained on count-based vectors, producing probabilistic topic distributions for each document. In parallel, Non-Negative Matrix Factorization (NMF) was applied to TF-IDF matrices, decomposing the corpus into latent topic components.

Both models generated five topics, which were interpreted by examining their highest-weighted keywords. The models consistently identified recurring themes concerning staff service, room quality, location and accessibility, general satisfaction, and resort facilities such as pools, beaches and food services.

3.5 Results and Evaluation

To quantify topic relevance, the dominant topic for each document was computed for both models and summarized in statistical output files. In the NMF model, room quality and related issues emerged as the most prevalent theme, dominating 25.5% of all reviews, followed by location and accessibility (22.6%) and resort amenities (20.1%). Topics related to staff service and overall experience (16.8%) and general praise (15.0%) were also frequently observed.

LDA produced a slightly different ranking: staff service and overall experience dominated 27.9% of reviews, followed by location (23.5%) and resort amenities (19.0%). Despite differences in ordering, the two models showed strong thematic overlap, suggesting that accommodation conditions, staff interactions, location factors and facility quality represent the central drivers of customer feedback.

The convergence between LDA and NMF increases confidence in the robustness of the extracted topics and demonstrates that combining multiple modeling techniques provides a more reliable understanding of unstructured text corpora.

3.6 Reflection and Learning Outcomes

Several challenges were encountered during implementation. Library compatibility issues and warnings related to Word2Vec on macOS required troubleshooting and package reinstallation. Parameter tuning for vectorizers and topic models also demanded iterative experimentation, particularly with respect to document-frequency thresholds, the number of topics and convergence settings.

From a personal perspective, the project strengthened practical skills in structuring end-to-end data analysis workflows, debugging environment issues, and interpreting unsupervised learning results critically rather than accepting outputs at face value. The comparison between different modeling approaches highlighted the importance of transparency and interpretability when developing systems intended to support public decision-making.

3.7 Limitations and Future Improvements

Although the TripAdvisor dataset served as a useful proxy for citizen complaints, it originates from the tourism domain and therefore cannot fully represent the diversity of topics present in municipal feedback. In a real-world deployment, multilingual data, demographic biases and uneven participation rates would require additional handling strategies.

Future improvements could include experimenting with transformer-based embeddings such as Sentence-BERT, dynamic topic modeling to track topic changes over time, and coherence metrics to systematically select the optimal number of topics. Incorporating metadata such as timestamps or user location could further enrich analyses and enable trend detection for emerging concerns.

3.8 Conclusion

This project demonstrates that NLP-based topic modeling pipelines can effectively summarize large volumes of unstructured feedback and provide decision-makers with quantitative insights into the most pressing concerns expressed by users. By combining careful preprocessing, multiple vectorization strategies and complementary topic modeling approaches, the analysis produced consistent and interpretable results, illustrating the potential of such systems for supporting data-driven governance.